

WEST BEND, WI, USA

WMO: 726413

Lat: 43.417N

Lon: 88.133W

Elev: 884

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04875

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.4	0.3	-15.1	2.5	-3.5	-9.3	3.4	2.2	23.4	20.3	20.7	20.1	7.2	280	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	88.2	72.5	84.4	70.6	82.0	69.0	75.1	83.2	73.3	81.8	71.8	79.7	10.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	124.0	80.2	70.5	116.0	77.7	69.7	112.7	77.0	39.0	83.2	37.5	81.5	36.1	79.5	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.6	16.7	DB	-10.6	92.4	5.0	3.2	-14.2	94.7	-17.2	96.6	-20.0	98.4	-23.6	100.7	(4)
			WB	-11.2	77.9	4.9	2.2	-14.8	79.5	-17.7	80.8	-20.4	82.0	-24.0	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	20.8	22.7	33.6	44.7	55.8	65.5	70.2	68.2	61.5	49.5	37.6	26.4	
	DBStd	19.52	11.78	11.07	11.06	9.09	8.55	6.89	5.97	5.43	7.68	8.69	9.42	10.53	
	HDD50	3683	906	764	524	210	38	0	0	6	119	384	732		
	HDD65	7229	1371	1183	975	610	308	75	18	28	156	488	821	1196	
	CDD50	2410	0	0	15	52	217	467	627	565	351	102	13	1	
	CDD65	483	0	0	1	3	22	92	180	128	51	6	0	0	
	CDH74	4289	0	0	7	60	252	867	1638	989	421	55	0	0	
	CDH80	1128	0	0	0	11	51	249	490	226	94	7	0	0	
Wind	WSAvg	6.8	8.1	7.8	7.4	8.2	7.0	5.8	5.2	4.7	5.3	6.9	7.5	7.6	
Precipitation	PrecAvg	32.0	1.2	1.0	2.0	2.9	2.9	3.5	3.8	3.7	4.2	2.7	2.4	1.6	
	PrecMax	41.4	3.8	3.1	5.5	4.5	6.5	7.2	10.3	7.4	10.5	5.7	6.8	3.9	
	PrecMin	22.6	0.1	0.1	0.3	0.9	0.4	1.4	1.1	0.4	0.2	0.2	0.2	0.0	
	PrecStd	5.1	0.8	0.7	1.2	1.0	1.4	1.7	1.9	1.7	2.8	1.5	1.5	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	54.8	72.6	81.2	85.7	90.9	93.4	89.9	87.6	80.8	67.2	57.0	
		MCWB	45.9	46.9	59.7	62.4	68.8	72.8	75.2	73.7	70.8	68.0	56.8	55.8	
	2%	DB	44.9	46.3	62.8	73.1	81.0	86.3	88.4	85.7	82.2	73.2	61.3	48.1	
		MCWB	40.6	41.4	53.5	58.7	65.3	70.5	73.5	71.2	68.9	63.2	54.2	45.5	
	5%	DB	39.0	41.2	54.9	66.5	76.5	82.0	84.5	81.9	79.2	69.7	56.5	44.6	
		MCWB	36.6	37.2	47.1	54.9	62.7	68.6	71.3	69.5	67.1	61.1	50.5	41.6	
	10%	DB	36.2	36.7	49.8	61.2	72.0	79.2	81.9	79.5	75.1	63.7	52.3	39.5	
		MCWB	34.6	34.0	43.5	50.4	60.1	66.3	69.8	67.8	65.3	56.9	47.3	36.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	48.1	61.0	67.3	71.1	76.6	78.2	76.4	74.2	70.9	60.1	55.2
			MCDB	47.7	53.7	70.6	74.0	81.6	85.9	87.8	84.9	82.2	77.8	63.5	57.4
2%		WB	40.3	42.4	55.4	60.2	67.8	73.3	75.3	74.2	71.7	66.8	55.7	46.1	
		MCDB	43.7	46.0	62.2	71.2	77.0	83.2	83.5	81.7	78.9	71.2	60.4	47.0	
5%		WB	37.1	38.1	49.5	56.5	65.2	71.0	73.5	72.3	69.6	62.2	51.6	41.7	
		MCDB	39.1	41.2	54.9	65.7	73.4	80.1	82.2	79.3	75.4	67.1	55.4	44.1	
10%		WB	34.3	34.4	43.5	51.5	62.3	68.6	72.1	70.3	67.5	57.8	47.1	37.5	
		MCDB	36.0	36.6	49.3	60.1	69.6	76.2	80.3	76.4	74.0	63.2	51.7	39.7	
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.7	16.1	18.6	20.0	19.8	20.0	19.3	20.3	17.4	14.5	12.1	
		MCDBR	15.7	18.6	23.4	28.8	26.1	23.0	21.8	21.8	23.5	24.4	21.3	15.6	
		MCWBR	13.2	14.0	15.6	16.5	14.2	11.5	10.0	10.6	12.1	14.4	15.1	13.4	
	5% WB	MCDBR	14.9	17.5	21.9	26.5	23.7	20.0	19.1	18.5	19.9	20.3	19.2	14.6	
		MCWBR	12.9	13.7	16.2	16.9	14.3	11.6	10.6	10.6	11.6	15.2	15.4	13.7	
Clear-Sky Solar Irradiance	taub	0.280	0.315	0.338	0.385	0.405	0.424	0.416	0.408	0.378	0.340	0.302	0.281		
	taud	2.394	2.284	2.360	2.271	2.262	2.255	2.277	2.292	2.363	2.468	2.509	2.443		
	Ebn at Noon	272	278	286	279	275	270	270	269	269	267	263	261		
	Edh at Noon	24	32	34	41	42	43	41	39	34	27	22	21		
All-Sky Solar Radiation	RadAvg	487	802	1164	1446	1702	1907	1948	1669	1350	859	543	385		
	RadStd	68	89	123	132	139	134	88	105	112	97	46	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page